Week 09 Question Bank

1. Explain the working procedure of the K-Means clustering algorithm with a suitable example.
2. Explain the Elbow Method and how it is used to select the optimal number of clusters.
3. Explain the role of inertia and the Dunn Index in evaluating the quality of clusters.
4. Explain dimensionality reduction and discuss why it is required in machine learning.
5. Differentiate between feature selection and feature extraction with suitable examples.
6. Explain Principal Component Analysis (PCA) and describe how it reduces the number of dimensions while preserving important information.
7. Differentiate between supervised learning and unsupervised learning with suitable examples.
8. Explain the different approaches of unsupervised learning: clustering, association, and dimensionality reduction.
9. Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) Initial cluster centres are: A1(2, 10), A4(5, 8) and A7(1, 2).
10. Explain MLOps and discuss its major benefits in machine learning projects.
11. Describe the different stages of a machine learning pipeline, from data ingestion to model deployment.
12. Differentiate between a transformer and an estimator in a machine learning pipeline with suitable examples.
13. Using the Mall Customers dataset, develop a Python program to determine the optimal number of clusters using the Elbow Method.

a) Load the dataset and select annual income and spending score.
b) Apply K-Means clustering for cluster counts ranging from 1 to 10.
c) Calculate and store the WCSS (Within-Cluster Sum of Squares) for each value of K.
d) Plot the Elbow graph with appropriate axis labels and title.
e) Identify a suitable value of K from the graph and justify your selection.
f) Fit the final K-Means model using the selected K and display the cluster centres.

14. Build Dimensionality Reduction [PCA] using iris Dataset .